

CHRIS TREES

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EXECUTIVE PROFILE

Senior technology executive with 40+ years architecting systems at the hardware-software boundary. Deep expertise spanning FPGA/ASIC design, distributed compute architecture, embedded platforms, AI systems, and edge infrastructure. Track record of founding and leading engineering ventures from concept through commercialization. Proven ability to translate complex R&D into deployable products across industries including financial services, healthcare, telecommunications, nuclear energy, and intelligent infrastructure. Currently focused on AI inference systems, voice AI, and bare metal orchestration for edge computing.

PROFESSIONAL EXPERIENCE

CAT9, LLC | Founder and Principal Researcher 2018 – Present | Remote

Founded research and consulting entity focused on edge network intelligence and next-generation AI infrastructure. Lead architecture decisions across AI, GPU computing, bare metal infrastructure, and DevOps automation.

- Architecting voice AI system (vapi.ai) that encodes 20 years of domain expert workflow knowledge into an AI assistant with human-in-the-loop oversight, enabling mobile-first business operations
- Directing bare metal infrastructure research using Digital Rebar Platform for CI/CD, SDN/SDS, and GPU-accelerated rendering pipelines
- Managing ASIC miner fleet operations and blockchain edge community infrastructure
- Advising startups and small businesses on technology architecture and legacy system modernization

Mail Services, LLC | Research Developer 2012 – 2018

Led evaluation and integration of Agile, CI/CD, QA testing, and system virtualization for production operations. Managed storage migration and maintained PHP-based production platform with SQL backend infrastructure.

OvalPath | Senior Software Engineer and Sales Engineer 2011 – 2012 | San Jose, CA and Cedar Falls, IA

Drove deployment of automated workforce tracking platform into commercial nuclear power station (Byron, IL) for the Radiation Protection division. Re-tooled mobile communications platform to fit regulated-environment field service workflows. Led QA test automation, build documentation, and customer-facing sales engineering. Company acquired by Curtiss-Wright.

CAT Services / Croft Utility Partners | Consultant and Founder 2009 – 2011

Founded community utility consulting practice focused on edge networking and rural infrastructure. Integrated power, water, sewer, and emergency communications systems for small rural communities.

Grasshorse LLC | Technology and Grant Consultant 2008 – 2009

Led technology for the ProtoMotion project — a large robotic stage system for stop motion animation production. Managed creative artist team through software production process. Secured state grant to commercialize robotics-assisted animation automation.

Vianeta Communications | Consultant 2005 – 2006

Data center deployment for healthcare voice transcription platform serving medical clinics. Stood up redundant Microsoft SQL database infrastructure. Company acquired by Spheris.

Pantera Systems | Co-Founder and Architect 2002

Co-founded with Ralph Aceves to commercialize distributed compute technology. Designed grid matrix architecture distributing storage, database, and compute algorithms across server clusters with indexed data structures for accelerated similarity lookups — a direct precursor to modern vector database patterns. Recruited Matt Wuebbeling (now VP at Nvidia) for marketing.

Fusion One | Sales Engineer 2001 | *San Jose, CA*

Integrated Levient redundant high-availability server infrastructure into multiple data centers. Led telecom and mobile cellular technology integrations. Developed scripted installation and network monitoring tools.

Monggo (Levient) | Systems Design 2000 – 2001

Developed one of the first remote desktop systems using WAP-enabled mobile devices. Built redundant high-availability server infrastructure deployed across multiple data centers. Created automation for UI testing and network health monitoring integrated into vendor NOCs.

FPGA Mortgage Data Accelerator | Hardware Architect 1999 – 2000

Designed and built large FPGA processing array accelerating weekly mortgage data analysis for derivative trading markets. Converted C/C++ routines into FPGA hardware, achieving 10,000x performance improvement against a 1,000x target — exceeding tape read bandwidth using only 2 of 12 FPGAs. Delivered loan package statistics 3-to-5 days ahead of Bloomberg. Acquired distribution rights to underlying IBM bubble sort patent technology.

Vertical Networks | Senior Systems Engineer 1997 – 2000

First hardware engineer hired. Led hardware and mechanical design for the Instant Office integrated LAN and telephony product from prototype through production. Recruited early-stage engineering talent.

Rapid City Communications | Consultant 1996 – 1997

Designed high-speed-density backplane. Collaborated on ASIC simulation work with hardware and software engineering teams.

SkyStream | Consultant 1996

FPGA design tool consulting engagement.

Sun Microsystems, Netra Group | Hardware Consultant 1995 – 1996

Led hardware engineering on the "Breadbox" project — an early Internet/Intranet appliance. Replaced Andy Bechtolsheim (Sun co-founder) as hardware engineer. Developed embedded control console, invented serial multiplexing system, integrated SunOS device drivers, designed multiple PCB assemblies with fabrication spanning multiple continents. Delivered on aggressive timeline navigating Sun internal processes as outside consultant.

General Magic, Inc. | Consultant 1994 – 1995

Maintained partner development systems and built mechanical and umbilical demo system for what is believed to be the first touch-based mobile phone prototype.

Hughes LAN Systems | Hardware Architect 1991 – 1993

Designed one of the first ATM networking boxes in the industry. Architected crossbar switching and cache management systems, balancing hardware, firmware, and software design domains for high-speed networking.

IKOS Systems | Sales Engineer 1990

Field sales engineer for ASIC simulation integrations across multiple silicon vendor libraries. Debugged silicon production issues for ASIC designers including OAK Technology.

The Engineering Dept. Inc. (TED) | Engineer and Board Member 1986 – 1989

Built PC Transporter ("Little Blue") — IBM emulator co-processor board for Apple II. Designed custom disk controller enabling cross-platform disk format compatibility. Pioneered early virtualization concepts, clone BIOS, and distributed callbacks. Collaborated with Lunar Design on award-winning industrial design. Worked alongside early Apple engineers including Wendell Sanders and Daniel Kottke.

Lockheed Martin | Engineer 1985 – 1986

First to introduce PLA and FPGAs to QA test fixture design. Built meta-script process converting vendor datasheets into functional timing simulation models. Inherited P387 video adaptor project 9 months behind schedule, delivered as first demo shown to customer.

EDUCATION

B.S. Electrical Engineering — Iowa State University

TECHNICAL PROFICIENCY

HDL/FPGA: VHDL, Verilog, Synopsys, Altera, Xilinx, QuickLogic | EDA/PCB: Mentor Graphics, OrCAD, Protel | Infrastructure: Docker, Linux, Digital Rebar, CI/CD | Languages: C, C++, Python, JavaScript, SQL | Methods: Agile, DevOps, Systems Design, Test Automation